

Kernel–IQ: Marginal Squeezing for Tail-Aware Covariance Estimation

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Abstract

Covariance estimation is central to portfolio optimization, yet many estimators either ignore tail structure or stabilize the empirical covariance matrix only after it has been formed. This paper introduces Kernel–IQ, a tail-aware covariance estimator that applies structured marginal squeezing to volatility-standardized returns before Gram aggregation. The core construction is spatial: observations are mapped into left-tail, body, right-tail, and central-exclusion regions, allowing asymmetric emphasis on downside and upside extremes while preserving positive semi-definiteness by construction. A temporal-spatial extension then applies a signed exponential row-weighting profile to the spatially squeezed design matrix before the Gram matrix is formed. In a long-only maximum-Sharpe portfolio study with monthly rebalancing and transaction costs, spatial marginal squeezing alone delivers strong out-of-sample portfolio-quality performance across repeated optimizer runs. The jointly tuned temporal-spatial specification improves this performance further, indicating that temporal weighting of the squeezed representation can complement the spatial tail-aware map. The results suggest that marginal squeezing provides a practical route to interpretable, tail-aware covariance estimation, preserving key IQ design principles while retaining the computational and structural simplicity of Gram-based PSD construction.

1 Introduction

Since [Markowitz \(1952\)](#), covariance estimation has remained central to portfolio construction, risk measurement, and asset allocation. Mean–variance portfolio theory requires a covariance matrix, but the sample covariance matrix is often noisy in finite samples and optimized portfolios can be highly sensitive to estimation error ([Michaud, 1989](#); [Best and Grauer, 1991](#); [Chopra and Ziemba, 1993](#); [Jagannathan and Ma, 2003](#)). Classical responses,

including linear and nonlinear shrinkage (Ledoit and Wolf, 2004a, 2017, 2022) and random-matrix cleaning methods (Laloux et al., 1999; Bun et al., 2017), stabilize the covariance matrix after the empirical second-moment object has been formed. These approaches can improve conditioning and portfolio behavior, but they leave unchanged the way individual observations enter covariance aggregation.

A different line of work constructs dependence estimates directly from co-movement events. Kendall’s tau (Kendall, 1938) and the Gerber statistic (Gerber et al., 2022) both depart from the raw covariance product as the primitive unit of measurement. Gerber’s Informational Quality framework, referred to here as original IQ, extends this constructive route by assigning structured spatial and temporal weights to co-movement evidence in the standardized bivariate return plane (Gerber et al., 2025). In that framework, squeezing refers to a controlled redistribution of evidential weight across regions of the return support, rather than an ex post repair of a covariance matrix. Subsequent work on Atomic-IQ develops a constructive covariance architecture based on positive-semidefinite event atoms and analytic spectral control, including an eigenvalue-floor mechanism (Abu Khalaf and Smyth, 2025).

Those IQ-style constructions are intentionally rich. Original IQ operates on pairwise co-movement locations in the bivariate return plane, while Atomic-IQ builds covariance estimates from PSD atomic components. That richer geometry is economically expressive, but it also makes the estimator more elaborate than may be necessary for every portfolio application. The question considered here is whether selected IQ design principles can be carried into a simpler, separable, marginal construction without losing the central economic idea: different regions of standardized return space may carry different informational quality for covariance estimation.

Kernel-IQ answers this question by moving the squeezing operation from the full pairwise return plane to the individual standardized marginals. Each asset return is mapped through a left-tail, body, right-tail, and central-exclusion transformation. The transformed observations are then assembled into a design matrix, and the dependence estimate is obtained from a Gram matrix. This gives a covariance estimator with four useful properties. First, the Gram construction preserves positive semi-definiteness. Second, the parameters remain interpretable: η_L , η_B , and η_R control left-tail, body, and right-tail emphasis, while δ_L and δ_R determine the tail boundaries. Third, bounded transformed entries control observation-level contributions before covariance aggregation. Fourth, the matrix-construction step is computationally light: once the marginal map has been applied, covariance construction is a single Gram product, a form for which standard numerical linear-algebra routines are highly optimized.

The paper also studies a temporal-spatial extension. In the original IQ and Atomic-IQ lineage, temporal parameters encode the idea that co-movement evidence can age and that recent evidence may be more or less informative than older evidence. In Kernel-IQ, the temporal parameter γ is implemented in a Gram-compatible way: after the spatial squeezing map has produced a design matrix Z , a signed exponential temporal profile weights the rows of Z before Gram aggregation. Positive γ emphasizes more recent squeezed observations, negative γ emphasizes the early part of the live window, and $\gamma = 0$ recovers the spatial-only

estimator.

The empirical setting is a long-only maximum-Sharpe portfolio problem over a diversified multi-asset universe, with monthly rebalancing and 10 basis points of transaction cost applied to turnover. Hyperparameters are selected in sample using Optuna (Akiba et al., 2019), and the full tuning-and-evaluation pipeline is repeated over 100 optimizer seeds. Two Kernel–IQ specifications are evaluated. The first is spatial-only, with $\gamma = 0$. The second tunes γ jointly with the spatial squeezing parameters in a deliberately bounded signed range. For transparency, the empirical section also reports an aggregate portfolio-quality rank. This rank summarizes ten out-of-sample performance, risk, downside, and implementation metrics, with the Sharpe ratio given double weight because the portfolio problem itself is a MaxSR allocation problem.

The results support two conclusions. First, spatial marginal squeezing alone is effective in the out-of-sample MaxSR experiment. Second, the temporal-spatial version improves the estimator further, with better risk-adjusted performance, drawdown behavior, and tail-risk characteristics than the spatial-only version. The advantage is not raw-return maximization. Higher-return shrinkage variants remain available, but they do so with higher realized volatility, deeper drawdowns, weaker tail-risk measures, and higher turnover. The Kernel–IQ result is therefore best interpreted as a portfolio-quality result: the estimator combines return, risk control, downside protection, and trading discipline in a way that is favorable for the MaxSR objective.

The remainder of the paper is organized as follows. Section 2 defines the spatial and temporal-spatial Kernel–IQ estimators. Section 3 describes the empirical design. Section 4 reports the MaxSR results and the nested comparison between the spatial-only and temporal-spatial specifications. Section 5 concludes. The appendices report the asset universe, implementation settings, tuning domains, optimized parameter summaries, benchmark definitions, full benchmark performance, and aggregate-rank summaries.

2 Kernel–IQ Construction

2.1 Spatial marginal squeezing

Let r_{it} denote the return of asset i at time t . At each estimation date, Kernel–IQ uses a live dependence window of length T and a longer volatility-scaling window of length $T^{(L)}$. The live window is used to construct local transformed co-movement evidence, while the longer volatility window stabilizes the units in which the squeezing thresholds are measured. This separation reduces the chance that the same short sample both defines the scale and determines the transformed dependence representation.

Let $r_{0,i}$ denote an asset-specific location estimate and let $\sigma_i^{(L)}$ denote the volatility estimate computed from the longer scaling window. The live-window standardized return is

$$x_{it} = \frac{r_{it} - r_{0,i}}{\sigma_i^{(L)}}. \quad (1)$$

At the matrix level, let $R \in \mathbb{R}^{T \times N}$ denote the live-window return matrix with entries $R_{ti} = r_{it}$, let

$$r_0 = (r_{0,1}, \dots, r_{0,N})^\top,$$

and define the diagonal long-window volatility matrix

$$D_\sigma^{(L)} = \text{diag}(\sigma_1^{(L)}, \dots, \sigma_N^{(L)}).$$

Then the standardized return matrix $X \in \mathbb{R}^{T \times N}$, with entries $X_{ti} = x_{it}$, may be written compactly as

$$X = (R - \mathbf{1}_T r_0^\top) (D_\sigma^{(L)})^{-1}, \quad (2)$$

where $\mathbf{1}_T$ is a T -dimensional vector of ones.

The thresholds $\delta_L > c \geq 0$ and $\delta_R > c \geq 0$ are measured in these standardized units. The parameter c defines a central exclusion region around zero, while δ_L and δ_R separate body observations from left-tail and right-tail observations. Figure 1 illustrates how these marginal regions replace the full bivariate IQ template.

The spatial Kernel-IQ map is

$$\phi_{\theta_S}(x) = \begin{cases} -\eta_L, & x < -\delta_L, \\ -\eta_B, & -\delta_L \leq x < -c, \\ 0, & |x| \leq c, \\ +\eta_B, & c < x \leq \delta_R, \\ +\eta_R, & x > \delta_R, \end{cases} \quad (3)$$

where

$$\theta_S = (\eta_L, \eta_B, \eta_R, \delta_L, \delta_R), \quad \eta_L, \eta_B, \eta_R \in [0, 1].$$

The parameter η_L controls left-tail emphasis, η_B controls non-excluded body emphasis, and η_R controls right-tail emphasis. The map is signed, bounded, and asymmetric across tails. It also removes very small standardized observations through the exclusion band $[-c, c]$.

Let $Z \in \mathbb{R}^{T \times N}$ be the transformed design matrix with entries

$$Z_{ti} = \phi_{\theta_S}(x_{it}). \quad (4)$$

Equivalently, if Φ_{θ_S} denotes elementwise application of the scalar map ϕ_{θ_S} to a matrix, then

$$Z = \Phi_{\theta_S}(X). \quad (5)$$

Thus the spatial transformation may be viewed as the sequence

$$R \longrightarrow X = (R - \mathbf{1}_T r_0^\top) (D_\sigma^{(L)})^{-1} \longrightarrow Z = \Phi_{\theta_S}(X).$$

The spatial Kernel-IQ Gram accumulator is

$$G_0 = Z^\top Z = \Phi_{\theta_S}(X)^\top \Phi_{\theta_S}(X). \quad (6)$$

The corresponding correlation matrix is obtained by diagonal normalization,

$$C_0 = D_G^{-1/2} G_0 D_G^{-1/2}, \quad D_G = \text{diag}(G_0), \quad (7)$$

where D_G denotes the diagonal matrix formed from the diagonal entries of G_0 . We use the usual numerical convention that assets with zero accumulator mass receive a unit diagonal and zero off-diagonal entries. Finally, the covariance estimator is

$$\hat{\Sigma}_0 = D_\sigma^{(L)} C_0 D_\sigma^{(L)}. \quad (8)$$

Since G_0 is a Gram matrix, it is positive semi-definite. Diagonal normalization and volatility rescaling preserve positive semi-definiteness when the accumulator diagonal is positive. The estimator therefore controls observation-level contributions before aggregation while producing a covariance matrix suitable for portfolio optimization.

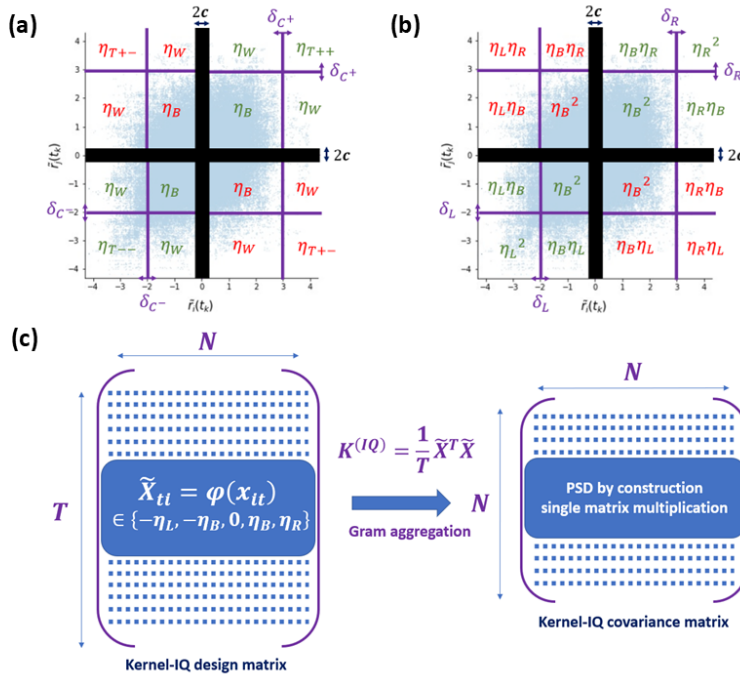


Figure 1: Conceptual progression from original IQ-style squeezing to Kernel-IQ. Panel (a) illustrates an IQ-style template defined on the bivariate return support. Panel (b) shows the induced bivariate weighting structure generated by separable marginal squeezing, with effective weights such as η_L^2 , η_R^2 , $\eta_L \eta_R$, and $\eta_B \eta_R$. Panel (c) shows the unweighted spatial Kernel-IQ design matrix and Gram aggregation step, corresponding to the $\gamma = 0$ construction. The temporal-spatial extension is defined separately in Section 2.2. The marginal construction preserves selected tail-aware design principles while obtaining positive semi-definiteness through Gram aggregation.

Although the transformation is defined marginally, the resulting covariance still encodes economically meaningful tail structure. Extreme same-sign observations contribute terms proportional to η_L^2 or η_R^2 , while opposite-sign extremes contribute terms proportional to $\eta_L\eta_R$. Body-tail interactions contribute products such as $\eta_B\eta_R$ or $\eta_B\eta_L$. Thus, Kernel-IQ does not reproduce the full non-separable geometry of original IQ, but it induces a structured bivariate weighting pattern through a simpler separable map. This induced structure is shown in Panel (b) of Figure 1.

2.2 Temporal-spatial extension

The temporal-spatial extension introduces a directional temporal weighting profile before the Gram matrix is formed. The spatial map ϕ_{θ_S} is applied to the standardized return matrix X , producing the squeezed design matrix

$$Z = \Phi_{\theta_S}(X).$$

The temporal parameter γ then weights the rows of Z in the Gram accumulator. Let A_γ denote the diagonal row-weighting matrix defined below. The order of operations is therefore

$$X \longrightarrow Z = \Phi_{\theta_S}(X) \longrightarrow Z^\top A_\gamma Z. \quad (9)$$

Index the live window by $t = 0, 1, \dots, T-1$, where $t = 0$ is the oldest observation and $t = T-1$ is the most recent observation. Define the directional distance

$$d_\gamma(t) = \begin{cases} T-1-t, & \gamma \geq 0, \\ t, & \gamma < 0. \end{cases} \quad (10)$$

With delay parameter $\epsilon \geq 0$, the raw temporal profile is

$$b_t(\gamma) = \exp\{-|\gamma|(d_\gamma(t) - \epsilon)_+\}, \quad (x)_+ = \max\{x, 0\}. \quad (11)$$

The parameter ϵ permits a delay before exponential decay begins away from the temporally emphasized edge of the live window. In the empirical implementation below, $\epsilon = 0$, so there is no such delay and the temporal profile is anchored directly to the live window. The profile is mass-normalized as

$$a_t(\gamma) = T \frac{b_t(\gamma)}{\sum_{s=0}^{T-1} b_s(\gamma)}, \quad (12)$$

so that $\sum_t a_t(\gamma) = T$. This normalization means that γ redistributes temporal emphasis within the live window without changing the total mass of the Gram accumulator. Positive values of γ emphasize more recent squeezed observations, negative values emphasize the early edge of the live window, and $\gamma = 0$ gives $a_t = 1$ for all t .

Let

$$A_\gamma = \text{diag}(a_0(\gamma), \dots, a_{T-1}(\gamma)). \quad (13)$$

The temporal-spatial accumulator is

$$G_\gamma = Z^\top A_\gamma Z = \Phi_{\theta_S}(X)^\top A_\gamma \Phi_{\theta_S}(X) = (A_\gamma^{1/2} Z)^\top (A_\gamma^{1/2} Z). \quad (14)$$

The same diagonal normalization and volatility rescaling then give

$$C_\gamma = D_{G,\gamma}^{-1/2} G_\gamma D_{G,\gamma}^{-1/2}, \quad \hat{\Sigma}_\gamma = D_\sigma^{(L)} C_\gamma D_\sigma^{(L)}, \quad (15)$$

where $D_{G,\gamma} = \text{diag}(G_\gamma)$ is the diagonal matrix formed from the diagonal entries of G_γ . This preserves positive semi-definiteness by the same Gram argument.

The temporal-spatial specification should not be interpreted as a post-hoc adjustment to a completed covariance matrix. Since γ is optimized jointly with $(\eta_L, \eta_B, \eta_R, \delta_L, \delta_R)$, the fitted estimator learns both which marginal return regions matter and how the corresponding squeezed observations should be temporally weighted. In that sense, the temporal and spatial components are complementary: the spatial map determines the event representation, while γ determines how that representation is accumulated over the live window.

3 Empirical Design

The empirical study evaluates Kernel-IQ in a long-only maximum-Sharpe portfolio problem using monthly data for a diversified multi-asset universe. The universe contains five equity indices, two fixed-income indices, listed real estate, gold, and a broad commodities index. The full asset list is reported in Appendix A. All methods use the same return and risk-free series, the same monthly rebalancing schedule, and the same transaction-cost assumption of 10 basis points applied to turnover. The portfolio problem is therefore held fixed while the covariance estimator varies.

The dependence window is determined by

$$T = \lceil qN \rceil, \quad (16)$$

with $q = 2.0$ and $N = 10$, so that $T = 20$. The volatility-scaling window is

$$T^{(L)} = mT, \quad (17)$$

with $m = 4$, yielding an 80-month scaling window. Returns are mean-centered, and the central exclusion width is fixed at $c = 0.05$. The design separates the short dependence window from the longer volatility-standardization window: the most recent 20 monthly returns define the squeezed co-movement representation, while the most recent 80 monthly returns define the volatility units used by c , δ_L , and δ_R . The intention is stability of scale combined with local adaptation of dependence.

Hyperparameters are selected over the in-sample evaluation period January 1995 to December 2004. The longer pre-history before January 1995 is used only to ensure that long-window volatility standardization is available at the first scored in-sample date. Out-of-

sample evaluation runs from January 2005 to December 2024. Kernel-IQ is tuned with 500 Optuna trials per seeded run using the annualized Sharpe ratio of monthly excess returns as the objective. The full pipeline is repeated for 100 optimizer seeds. The optimized benchmark variants, GS_{opt} and SRE_{opt} , are also re-estimated within their restricted parameter spaces: GS_{opt} tunes the Gerber threshold over $[0, 1]$, while SRE_{opt} tunes the residual-eigenvalue shrinkage parameter α over $[0, 1]$. The remaining benchmark estimators retain fixed specifications. Appendix E provides the full benchmark definitions and references.

Two Kernel-IQ specifications are reported. The first is the spatial-only estimator with $\gamma = 0$. This specification tests whether marginal tail-aware squeezing alone is useful. The second is the temporal-spatial estimator in which γ is tuned jointly with the spatial parameters over the signed range $[-0.10, 0.10]$ using a step size of 0.02. In both cases, standardized returns are first allocated to tail, body, or exclusion regions by the spatial squeezing map; only after this allocation does the temporal-spatial specification apply row weights to the squeezed design matrix. Both specifications use the same asset universe, data split, rebalancing rule, transaction-cost assumption, and benchmark menu.

For each seeded run, we record annualized return, cumulative return, annualized volatility, maximum drawdown, monthly 95% value-at-risk, monthly 95% expected shortfall, Sharpe ratio, Sortino ratio, Calmar ratio, and turnover. We also compute per-metric ranks and an aggregate-rank score. Ranks are oriented so that rank 1 always denotes the preferred outcome: higher is better for return, cumulative return, Sharpe, Sortino, and Calmar; lower is better for volatility, drawdown magnitude, VaR loss magnitude, expected-shortfall loss magnitude, and turnover. For drawdown, VaR, and expected shortfall, ranks are based on loss magnitudes, so less severe losses are preferred even though the tabulated values are reported with negative signs. Because the portfolio problem is explicitly a maximum-Sharpe problem, the aggregate rank gives double weight to the Sharpe ratio and single weight to each remaining metric. The theoretical lower bound is therefore 11.

The 100-seed ensemble is intended to assess stability with respect to optimizer randomization, not to create independent market samples. Each seed produces a fresh hyperparameter search within a fixed historical design. The ensemble therefore asks whether the tuned estimator’s performance is stable across distinct optimizer trajectories, reducing the possibility that the reported evidence is driven by a single favorable parameterization.

4 Results

4.1 Spatial-only and temporal-spatial performance

Table 1 reports mean out-of-sample performance across the 100 seeded MaxSR runs. The table includes both Kernel-IQ specifications and four selected comparators. These comparators are included in the main text because they are especially informative: GS_{opt} represents the tuned thresholded-concordance benchmark; LS1, LS2, and LS3 are leading linear-shrinkage alternatives, with LS1 and LS2 particularly useful because they generate higher raw returns than Kernel-IQ while carrying materially greater risk. The full benchmark menu and full

comparator results are reported in Appendices E and F.

The spatial-only estimator already gives a strong out-of-sample portfolio-quality result. With $\gamma = 0$, Kernel-IQ obtains annualized volatility of 8.98%, maximum drawdown of -29.66%, monthly 95% VaR of -3.59%, and turnover of 2.41. These figures compare favorably with GS_{opt} and the selected shrinkage comparators on risk, downside, and turnover dimensions, even though LS1 and LS2 generate higher raw returns.

The temporal-spatial version improves the central MaxSR evidence. With γ tuned jointly with the spatial parameters, Kernel-IQ raises annualized return to 6.39% while reducing annualized volatility to 8.86%. It also improves maximum drawdown to -27.27%, monthly 95% VaR to -3.10%, and cumulative return to 231.14%. Turnover rises from 2.41 to 2.53 relative to the spatial-only version, but remains materially below the selected comparator group.

Table 1: Mean out-of-sample portfolio metrics across 100 optimizer seeds

Metric	Kernel-IQ		GS_{opt}	LS2	LS3	LS1
	$\gamma = 0$	γ tuned				
Annualized Return (%)	6.05	6.39	5.63	8.04	5.62	7.15
Annualized STD (%)	8.98	8.86	9.38	13.98	9.88	12.02
Sharpe Ratio	0.497	0.542	0.428	0.460	0.410	0.460
Sortino Ratio	0.566	0.583	0.539	0.580	0.500	0.570
Calmar Ratio	0.206	0.238	0.182	0.190	0.170	0.200
Cumulative Return (%)	208.45	231.14	181.55	307.47	178.73	259.58
Maximum Drawdown (%)	-29.66	-27.27	-30.63	-41.75	-32.13	-36.53
Monthly 95% VaR (%)	-3.59	-3.10	-4.04	-6.46	-4.01	-4.83
Monthly 95% Expected Shortfall (%)	-5.97	-5.91	-6.35	-9.34	-6.94	-8.36
Turnover	2.41	2.53	3.82	4.07	3.62	3.45

The comparison with LS2 is informative. LS2 produces the highest annualized return and cumulative return in Table 1, but it does so with substantially higher volatility, deeper drawdowns, weaker VaR and expected-shortfall behavior, and higher turnover. Kernel-IQ therefore does not win by taking the most risk. Its advantage lies in the balance between return, risk-adjusted performance, downside protection, and trading intensity.

4.2 Spatial-only versus temporal-spatial specification comparison

Table 2 isolates the change from the spatial-only Kernel-IQ specification to the temporal-spatial specification. The tuned- γ version improves nine of the ten reported portfolio-quality dimensions if turnover is treated as a cost. The increase in turnover is modest and does not overturn the overall trading-intensity advantage of Kernel-IQ relative to the principal comparators.

The aggregate-rank evidence reinforces this specification comparison. Table 3 reports a single joint ranking field in which both Kernel-IQ specifications and the benchmark estimators are ranked together under the same MaxSR aggregate profile. The temporal-spatial estimator has the best mean aggregate rank, 29.69, followed by the spatial-only estimator at 45.48. The next best non-Kernel-IQ method is GS1 with a mean aggregate rank of 84.73.

Table 2: Spatial-only versus temporal-spatial Kernel-IQ specification comparison

Metric	$\gamma = 0$	γ tuned	Change
Annualized Return (%)	6.05	6.39	0.34
Annualized STD (%)	8.98	8.86	-0.12
Sharpe Ratio	0.497	0.542	0.045
Sortino Ratio	0.566	0.583	0.017
Calmar Ratio	0.206	0.238	0.032
Cumulative Return (%)	208.45	231.14	22.69
Maximum Drawdown (%)	-29.66	-27.27	2.40
Monthly 95% VaR (%)	-3.59	-3.10	0.49
Monthly 95% Expected Shortfall (%)	-5.97	-5.91	0.05
Turnover	2.41	2.53	0.12

In paired seed comparisons between the two Kernel-IQ specifications, the temporal-spatial version has a better aggregate rank than the spatial-only version in 65 of the 100 runs. This should not be read as independent-sample statistical inference, but it is strong evidence that the temporal-spatial specification is not merely a single favorable optimizer realization.

Table 3: Joint aggregate-rank summary under the MaxSR ranking profile

Method	Mean aggregate rank	Median
Kernel-IQ (γ tuned)	29.69	24.50
Kernel-IQ ($\gamma = 0$)	45.48	48.00
GS1	84.73	85.00
LS1	86.83	87.00
NLS8	95.07	95.00
NLS6	95.94	96.00
GS _{opt}	97.18	98.00
GS3	101.74	101.00
LS5	102.29	102.00
LS2	106.16	106.00

4.3 Parameter means and mechanism evidence

Across the 100 temporal-spatial seeded runs, the mean selected temporal parameter is $\bar{\gamma} = 0.086$. Since γ is optimized over the bounded signed range $[-0.10, 0.10]$, this indicates a persistent preference for a positive temporal profile within the permitted range. In this implementation, positive γ gives more temporal mass to recent squeezed observations before the Gram matrix is formed. The result should therefore be interpreted as evidence that modest recency emphasis is useful in this MaxSR experiment, not as a claim that the particular bound is globally optimal.

The spatial parameter means also change when γ is enabled. The spatial-only fit uses a high left-tail weight and a very small body weight, with mean values $\eta_L = 0.836$ and $\eta_B = 0.015$. The temporal-spatial fit reduces the left-tail weight to $\eta_L = 0.614$, raises the body weight to $\eta_B = 0.148$, and raises the right-tail weight to $\eta_R = 0.389$. This change is important. It shows that the tuned- γ version is not simply the spatial-only estimator plus an extra row-weighting parameter. Because γ and the spatial parameters are optimized jointly, the temporal-spatial estimator learns a different balance between regional squeezing and temporal evidence accumulation.

Figure 2 separates active-region occupancy from the learned squeezing weights. In the spatial-only model, 91.6% of active in-sample observations lie in the body region, while only 3.1% and 5.3% lie in the left and right regions respectively. In the temporal-spatial model, the corresponding active-region shares are 8.3% left, 86.9% body, and 4.8% right. The normalized squeezing weights move in the opposite direction: both specifications give much less raw weight to the body than its empirical frequency would imply. The estimator therefore does not merely reproduce empirical region occupancy.

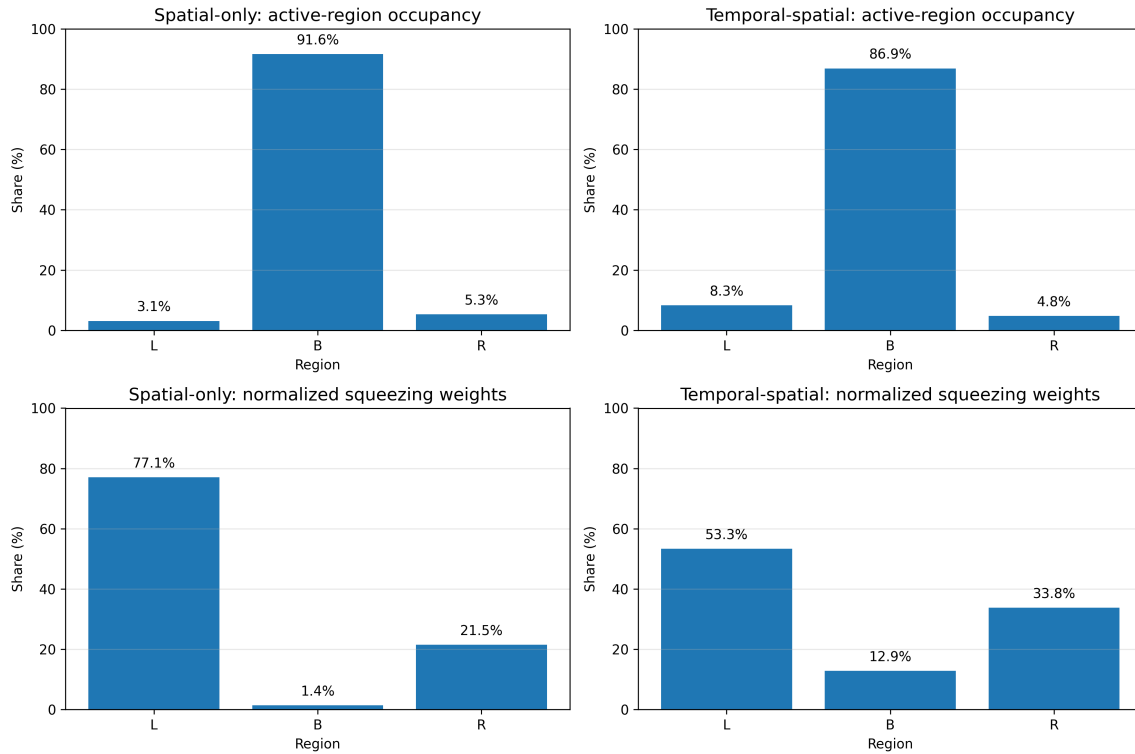


Figure 2: Active-region occupancy and normalized squeezing weights for spatial-only and temporal-spatial Kernel-IQ. The upper panels report average in-sample active-region occupancy shares for left-tail (L), body (B), and right-tail (R) observations. The lower panels report normalized mean squeezing weights based on η_L , η_B , and η_R . Both specifications assign substantially less raw squeezing weight to the body region than its empirical occupancy would imply.

Figure 3 combines occupancy, squeezing weights, and, for the temporal-spatial model, the fitted temporal row weights. In the spatial-only model, effective emphasis is 55.3% left,

23.0% body, and 21.8% right in sample, with a similar out-of-sample pattern of 54.4%, 24.5%, and 21.0%. In the temporal-spatial model, effective emphasis is 49.5% left, 32.5% body, and 18.0% right in sample, again remaining stable out of sample at 48.6%, 33.9%, and 17.5%. Kernel-IQ therefore reallocates evidential emphasis away from the central mass and toward less frequent regions, especially the downside region, while the temporal-spatial specification does so with less extreme body suppression than the spatial-only version.

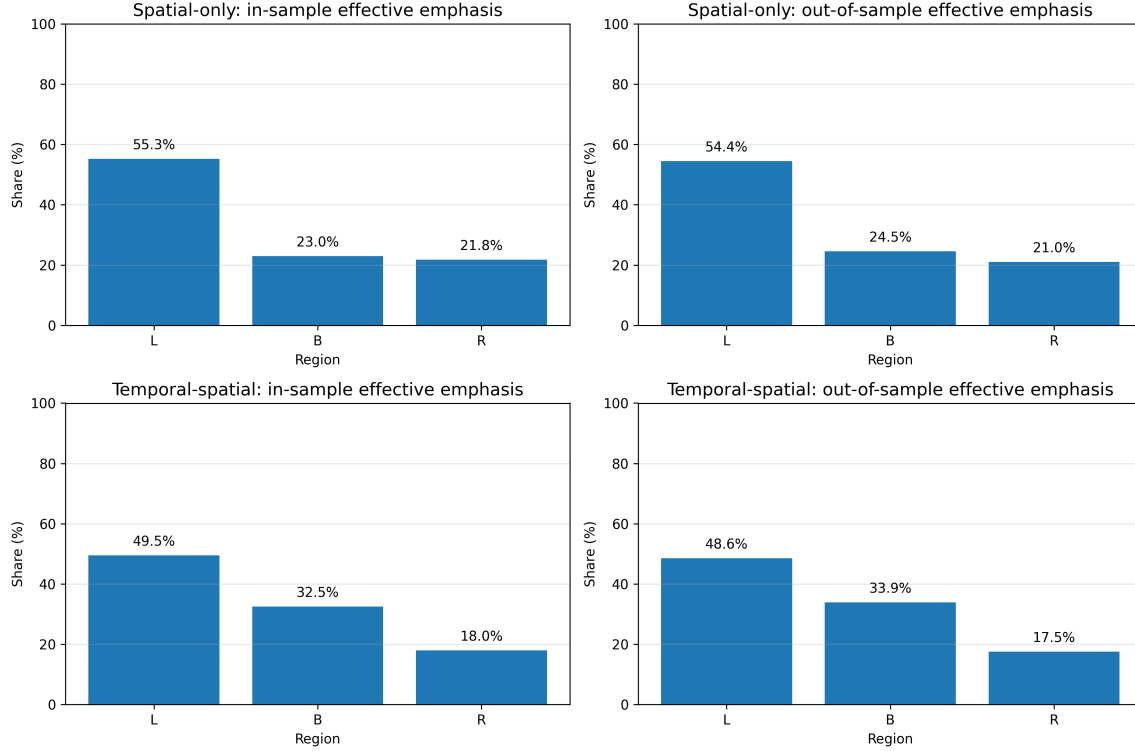


Figure 3: Effective region emphasis for spatial-only and temporal-spatial Kernel-IQ. Each panel combines region occupancy with the learned squeezing weights and normalizes across active regions. For the temporal-spatial specification, the calculation also incorporates the signed temporal row weights applied before Gram aggregation. The comparison shows that effective emphasis remains stable from the in-sample tuning period to the out-of-sample evaluation period.

5 Conclusion

Kernel-IQ shows that selected IQ design principles can be implemented through a simple marginal covariance construction. By relocating squeezing from the full pairwise return support to standardized marginal returns, the estimator retains interpretable tail asymmetry while producing a PSD covariance matrix through Gram aggregation. The construction is also computationally light in the matrix-construction step, because the non-linear part of the estimator is a separable marginal map followed by a single optimized Gram product. This creates a practical bridge between white-box tail-aware covariance estimation and matrix-based portfolio implementation.

The empirical results support two claims. First, spatial marginal squeezing alone is useful: the $\gamma = 0$ version of Kernel-IQ performs strongly in the 100-seed MaxSR experiment and remains highly competitive across repeated optimizer runs. Second, temporal weighting of the spatially squeezed representation improves the estimator further: in the joint ranking field, the temporal-spatial version improves the mean aggregate rank from 45.48 for the spatial-only version to 29.69 and remains the leading specification across the seeded diagnostic. The substantive interpretation is not that Kernel-IQ maximizes raw return, but that it produces a favorable out-of-sample balance between return, volatility, drawdown, tail risk, and turnover.

The findings should be read with appropriate scope. The study uses one diversified multi-asset universe, one monthly frequency, one long-only MaxSR objective, and one historical sample period. The optimizer-seed ensemble measures tuning stability rather than independent market replication. The results do not claim that Kernel-IQ is a universal replacement for original IQ, Atomic-IQ, shrinkage, random-matrix methods, or Gerber-style estimators. They show instead that a marginal, PSD-preserving implementation of tail-aware squeezing can perform strongly in a realistic portfolio problem, and that a signed temporal row-weighting profile can add useful flexibility.

Future work can examine larger asset universes, alternative portfolio objectives, sensitivity to the temporal bound on γ , direct comparisons with original IQ and Atomic-IQ implementations, and variants in which temporal preconditioning is applied before the spatial squeezing map.

A Asset Universe

The empirical study uses a diversified multi-asset universe comprising five equity indices, two fixed-income indices, listed real estate, gold, and a broad commodities index. The asset universe is:

1. S&P 500 Index (SPX),
2. Russell 2000 Index (RTY),
3. MSCI EAFE Index (MXEA),
4. MSCI Emerging Markets Index (MXEF),
5. Nasdaq-100 Index (NDX),
6. Bloomberg Barclays U.S. Aggregate Bond Index (LBSTRUU),
7. Bloomberg Barclays U.S. Corporate High Yield Bond Index (LF98TRUU),
8. FTSE NAREIT All Equity REITs Index (FNERTR),
9. Gold (XAU),
10. S&P GSCI Commodity Index (SPGSCI).

The sample spans January 1988 to December 2024. Monthly asset-return series and the risk-free series were obtained from Bloomberg Terminal. The in-sample tuning evaluation period is January 1995 to December 2004, and the out-of-sample evaluation period is January 2005 to December 2024.

B Backtest Protocol and Fixed Design Choices

All covariance estimators are embedded in the same long-only maximum-Sharpe portfolio framework, use the same return and risk-free series, are rebalanced monthly, and are assessed under the same transaction-cost assumption of 10 basis points applied to turnover. Table 4 summarizes the key fixed design choices.

Table 4: Kernel-IQ specification

Quantity	Value / status	Description
N	10	Number of assets in the portfolio universe
q	2.0	Ratio determining the live dependence-window length
T	20	Live dependence-window length, $T = \lceil qN \rceil$
m	4	Multiplier for the long volatility-scaling window
$T^{(L)}$	80	Long volatility-scaling window, $T^{(L)} = mT$
$r_{0,i}$	sample mean	Asset-specific centering location
c	0.05	Half-width of the central exclusion region in standardized units
η_L, η_B, η_R	tuned	Left-tail, body, and right-tail squeezing weights
δ_L, δ_R	tuned	Left and right tail thresholds
γ	fixed or tuned	Fixed at 0 in the spatial-only model; tuned in the temporal-spatial model
Optuna trials	500 per Kernel-IQ run	Hyperparameter-search budget for Kernel-IQ
Optimizer seeds	100	Repeated tuning runs
In-sample period	Jan 1995 to Dec 2004	Hyperparameter-selection evaluation period
Out-of-sample period	Jan 2005 to Dec 2024	Performance-evaluation period
Transaction costs	10 bps	Applied to turnover at each rebalance

The fixed values $q = 2$, $m = 4$, and $c = 0.05$ are design choices rather than universal recommendations. The value $q = 2$ gives a short 20-month dependence window in this multi-asset setting, a natural choice for monthly returns when the objective is local covariance adaptation rather than long-run parameter estimation. The multiplier $m = 4$ separates the longer volatility-standardization window from the shorter co-movement construction window,

stabilizing the scale used by the thresholds. The exclusion width $c = 0.05$ suppresses very small standardized observations, where sign information is least informative.

C Tuning Design and Search Domains

Table 5 separates fixed quantities from tuned quantities. The spatial-only and temporal-spatial specifications share the same spatial search domain; the latter additionally tunes γ .

Table 5: Tuning design and search domains

Quantity	Status / domain	Role
η_L	tuned, $[0, 1]$	Downside-tail squeezing weight
η_B	tuned, $[0, 1]$	Body-region squeezing weight
η_R	tuned, $[0, 1]$	Upside-tail squeezing weight
δ_L	tuned, $\{1.0, 1.2, \dots, 2.0\}$	Left-tail threshold in standardized units
δ_R	tuned, $\{1.0, 1.2, \dots, 2.0\}$	Right-tail threshold in standardized units
γ	fixed at 0 or tuned in $[-0.10, 0.10]$ with step 0.02	Signed temporal row-weighting parameter
ϵ	fixed at 0	Delay before temporal decay begins
q	fixed at 2.0	Sets live dependence-window length
m	fixed at 4	Sets volatility-scaling window relative to live window
c	fixed at 0.05	Sets central-exclusion half-width

D Optimized Kernel–IQ Parameter Values

Table 6 reports mean and standard deviation of the optimized Kernel–IQ parameter values across the 100 optimizer-seed runs.

Table 6: Optimized Kernel-IQ parameter summaries across 100 seeds

Parameter	Spatial-only ($\gamma = 0$)		Temporal-spatial (γ tuned)	
	Mean	SD	Mean	SD
η_L	0.836	0.116	0.614	0.293
η_B	0.015	0.026	0.148	0.289
η_R	0.234	0.090	0.389	0.302
δ_L	1.994	0.034	1.478	0.299
δ_R	1.692	0.242	1.724	0.192
γ	0.000	0.000	0.086	0.036

The parameter summaries are consistent with the mechanism evidence in Figures 2 and 3. The spatial-only estimator uses very low body weighting and strong left-tail emphasis. Once temporal weighting is enabled, the fitted spatial map becomes less extreme in the body and right-tail dimensions while the mean γ becomes positive.

The standard deviations in Table 6 also show that the spatial-only specification has a more concentrated fitted-parameter distribution, particularly for η_L , η_R , and δ_L . This should not be interpreted as proving more stable portfolio outcomes by itself, since parameter dispersion and performance stability are not the same object. The more relevant point is that the temporal-spatial specification explores a broader joint parameter space on the same search budget, yet still delivers stable and improved out-of-sample aggregate performance in the seeded optimizer ensemble.

E Benchmark Estimators and Tuning

The benchmark menu spans historical covariance, linear shrinkage, nonlinear shrinkage, random-matrix cleaning, and Gerber-style concordance estimators. Table 7 summarizes the full comparison set and gives the primary methodological reference for each family.

Table 7: Benchmark estimators considered in the full comparison set

Technique	Description
HC	Historical sample covariance matrix.
LS1	Linear shrinkage toward a one-parameter target in which all variances are equal and all covariances are zero (Ledoit and Wolf, 2004b).
LS2	Linear shrinkage toward a two-parameter target in which all variances are equal and all covariances are equal (Ledoit, 1995).
LS3	Linear shrinkage toward a constant-correlation target that preserves the sample variances and sets all correlations equal (Ledoit and Wolf, 2004a).
LS4	Linear shrinkage toward a diagonal target that preserves sample variances and sets covariances to zero (Ledoit, 1995).

Technique	Description
LS5	Linear shrinkage toward a one-factor market-model target (Ledoit and Wolf, 2003).
NLS6	Geometric-inverse shrinkage under symmetrized Kullback–Leibler loss, combining the linear-inverse and quadratic-inverse perspectives (Ledoit and Wolf, 2022).
NLS7	Linear-inverse shrinkage derived under Stein’s loss (Ledoit and Wolf, 2022; Stein, 1986).
NLS8	Quadratic-inverse shrinkage derived under Frobenius, inverse Stein, and minimum-variance loss criteria (Ledoit and Wolf, 2022).
CRE	Constant residual eigenvalue random-matrix cleaning: Marchenko–Pastur analysis is used to identify noise-associated eigenvalues, which are replaced by their average while preserving the trace (Marchenko and Pastur, 1967; Laloux et al., 1999; Bun et al., 2017; Lopez de Prado, 2020).
SRE	Shrinkage of residual eigenvalues random-matrix cleaning: the signal subspace is retained while the residual eigenvalue component is shrunk toward its diagonal form (Lopez de Prado, 2020). The fixed SRE benchmark uses $\alpha = 0.1$ and bandwidth $h = 0.01$.
GS1	Gerber Statistic with threshold $\theta = 0.5$ (Gerber et al., 2022).
GS2	Gerber Statistic with threshold $\theta = 0.7$ (Gerber et al., 2022).
GS3	Gerber Statistic with threshold $\theta = 0.9$ (Gerber et al., 2022).
GS _{opt}	Tuned Gerber Statistic variant. The threshold θ is selected in sample over $[0, 1]$ using the same MaxSR objective and 100 Optuna trials.
SRE _{opt}	Tuned SRE variant. The residual-eigenvalue shrinkage parameter α is selected in sample over $[0, 1]$ using the same MaxSR objective and 100 Optuna trials; bandwidth remains fixed at $h = 0.01$.
Kernel–IQ	Proposed marginal-squeezing covariance estimator, with either $\gamma = 0$ or tuned γ .

F Full Out-of-Sample Benchmark Performance

Tables 8 and 9 report the full set of mean out-of-sample benchmark metrics. Both Kernel–IQ specifications are included, so that the spatial-only and temporal-spatial versions can be compared against the full benchmark field.

Table 8: Full benchmark performance: return and risk-adjusted metrics

Method	Ann. Ret. (%)	Cum. Ret. (%)	Ann. STD (%)	Sharpe	Sortino	Calmar
Kernel-IQ ($\gamma = 0$)	6.05	208.45	8.98	0.497	0.566	0.206
Kernel-IQ (γ tuned)	6.39	231.14	8.86	0.542	0.583	0.238
CRE	5.07	152.48	9.31	0.370	0.450	0.160
GS1	5.31	166.33	9.01	0.410	0.500	0.200
GS2	4.67	135.05	8.82	0.350	0.410	0.160
GS3	4.87	146.68	8.39	0.390	0.470	0.170
GS _{opt}	5.63	181.55	9.38	0.428	0.539	0.182
HC	5.42	170.32	9.41	0.400	0.500	0.180
LS1	7.15	259.58	12.02	0.460	0.570	0.200
LS2	8.04	307.47	13.98	0.460	0.580	0.190
LS3	5.62	178.73	9.88	0.410	0.500	0.170
LS4	4.91	145.99	8.96	0.370	0.450	0.160
LS5	5.23	161.61	9.12	0.400	0.480	0.170
NLS6	5.93	194.47	10.15	0.430	0.530	0.200
NLS7	5.85	190.21	10.09	0.420	0.520	0.190
NLS8	6.00	198.37	10.20	0.430	0.530	0.200
SRE	5.22	160.67	9.21	0.390	0.480	0.170
SRE _{opt}	5.22	160.51	9.20	0.390	0.470	0.170

Table 9: Full benchmark performance: downside and implementation metrics

Method	Max. Drawdown (%)	Monthly 95% VaR (%)	Monthly 95% ES (%)	Turnover
Kernel-IQ ($\gamma = 0$)	-29.66	-3.59	-5.97	2.41
Kernel-IQ (γ tuned)	-27.27	-3.10	-5.91	2.53
CRE	-32.10	-3.42	-6.48	3.67
GS1	-27.22	-4.14	-6.31	3.68
GS2	-29.32	-3.80	-6.32	3.68
GS3	-29.36	-3.62	-5.81	3.53
GS _{opt}	-30.63	-4.04	-6.35	3.82
HC	-30.50	-4.01	-6.51	3.71
LS1	-36.53	-4.83	-8.36	3.45
LS2	-41.75	-6.46	-9.34	4.07
LS3	-32.13	-4.01	-6.94	3.62
LS4	-30.01	-4.01	-6.27	3.44
LS5	-31.45	-3.11	-6.31	3.58
NLS6	-30.22	-4.48	-7.08	3.91
NLS7	-30.25	-4.44	-7.05	3.91
NLS8	-30.21	-4.49	-7.12	3.91
SRE	-30.55	-3.47	-6.43	3.69
SRE _{opt}	-30.54	-3.57	-6.43	3.72

G Full Aggregate-Rank Summary

Table 10 reports the full joint aggregate-rank summary for both Kernel-IQ specifications and all benchmark estimators. Lower aggregate rank is better. The aggregate rank is based on ten metrics, with Sharpe ratio double-weighted, so the theoretical lower bound is 11. Both Kernel-IQ specifications are included in the same ranking field.

Table 10: Full joint aggregate-rank summary across 100 optimizer seeds

Method	Mean aggregate rank	Median
Kernel-IQ (γ tuned)	29.69	24.50
Kernel-IQ ($\gamma = 0$)	45.48	48.00
GS1	84.73	85.00
LS1	86.83	87.00
NLS8	95.07	95.00
NLS6	95.94	96.00
GS _{opt}	97.18	98.00
GS3	101.74	101.00
LS5	102.29	102.00
LS2	106.16	106.00
NLS7	107.85	108.00
LS3	113.26	113.00
HC	113.35	113.00
SRE	118.35	118.50
LS4	121.59	122.00
SRE _{opt}	122.63	123.00
GS2	135.56	136.00
CRE	140.35	140.00

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